

8 (a) (i) $\phi = B_H A$
 $= (5.2 \times 10^{-5} \cos 70^\circ)(0.80 \times 0.55)$
 $= 7.8254 \times 10^{-6} = 7.83 \times 10^{-6} \text{ Wb}$

- (ii) 1. When the window is opening, the aluminium frame cuts the Earth's magnetic field lines and the magnetic flux through the window decreases. This causes a rate of change of magnetic flux linkage in the frame.

By Faraday's law, there will be an induced e.m.f. in the frame.

Since the frame is a closed loop, this induced e.m.f. produces an induced current in the frame.

By Lenz's law, the direction of the induced current is such as to produce an effect to oppose the decrease in the magnetic flux. Current will flow from ADCB to increase the magnetic flux.

2. $\Delta\phi = \phi_f - \phi_i$
 $= 0 - 7.83 \times 10^{-6}$
 $= -7.83 \times 10^{-6} \text{ Wb}$

3. $|E| = \left| -\frac{\Delta\phi}{\Delta t} \right|$
 $= \left| -\frac{-7.83 \times 10^{-6}}{0.30} \right|$
 $= 2.61 \times 10^{-5} \text{ V}$

- (b) (i) Since the velocity of the charged particles is perpendicular to the magnetic field, the magnetic force acting on the charged particles is always perpendicular to their velocity. The magnetic force only changes the direction of the velocity while the speed remains constant.

As the magnitude of the magnetic force is constant, it points towards a fixed point, causing the particles to move in a circular motion about the fixed point with a constant radius.

- (ii) The magnetic force acting on the particle provides the centripetal force.

$$Bqv = m \frac{v^2}{r}$$

$$r = \frac{mv}{Bq}$$

Since the particles travel at the same speed in the field of the same magnetic flux density,

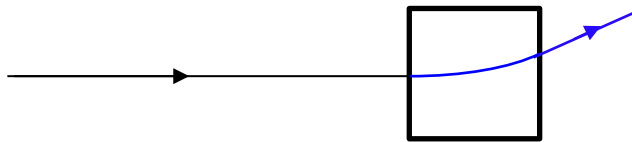
$$\frac{r_\alpha}{r_e} = \frac{m_\alpha q_e}{m_e q_\alpha}$$

$$= \frac{(4 \times 1.66 \times 10^{-27})}{9.11 \times 10^{-31}} \times \frac{1}{2}$$

$$= 3644.3$$

$$r_\alpha = 3.6 \times 10^3 r_e \quad (\text{shown})$$

(iii)



*Slight curve (due to large radius) in B field.

*Straight path after it exits B field from the right edge of the field.

(iv)

$$\begin{aligned}
 1. \quad R &= \frac{mv}{Bq} \\
 &= \frac{(9.11 \times 10^{-31})(1.2 \times 10^6)}{(0.70 \times 10^{-3})(1.60 \times 10^{-19})} \\
 &= 9.7607 \times 10^{-3} = 9.8 \text{ mm} \quad (\text{shown})
 \end{aligned}$$

$$\begin{aligned}
 2. \quad T &= \frac{2\pi R}{v} \\
 &= \frac{2\pi(9.8 \times 10^{-3})}{1.2 \times 10^6} \\
 &= 5.131 \times 10^{-8} = 5.13 \times 10^{-8} \text{ s}
 \end{aligned}$$

3. The electron moves in a circular path in a clockwise direction (viewed from top).

From 0 ms to 1.2 ms, the radius of curvature of its path decreases and increases from 1.2 ms to 3.2 ms.

After 3.2 ms, the direction of its circular path will change to an anti-clockwise direction.

From 3.2 ms to 4.0 ms, the radius of curvature of its path decreases.

4. State any time from 2.4 ms to just before 3.2 ms (from $B = 0.70 \text{ mT}$ and decreasing to zero).

As the magnetic flux density of the field decreases, the radius of curvature of the electron's path increases and leaves the magnetic field due to the high speed of the electron.

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0 (c) The maximum kinetic energy of the photoelectrons is dependent only on the frequency